MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ISOLATION MODULE, DIGITAL, KEY CODE EMF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic isolation module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0044 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EMF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/129-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through F shall be performed in any order.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Omitted tests. This module shall not be subjected to operating temperature tests, thermal shock test, life test, or maximum stress tests.

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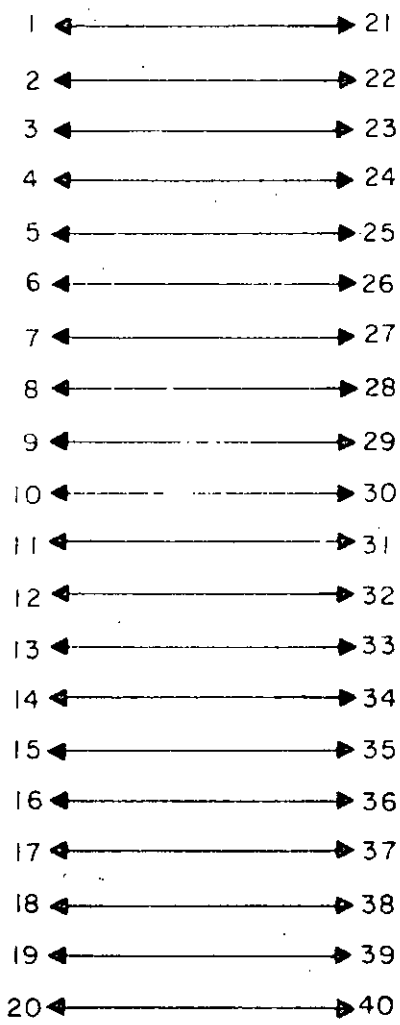


FIGURE 1. EMF function diagram, isolation module.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Module continuity	II	X	0	2	0	2	0	2	0	2	Ω
B1-B20	Module integrity	III	X	10	∞	10	∞	10	∞	10	∞	MΩ
C1-C20	Jumper resistance check	IV	S	0	50	0	50	0	50	0	50	mV
D1-D20	Insulation resistance	V	S	10	∞	10	∞	10	∞	10	∞	MΩ
E1-E20	Capacitance	VI	S	0	10	0	10	0	10	0	10	pF
F	Frame integrity	VII	X	10	∞	10	∞	10	∞	10	∞	MΩ

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/129 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658639B dated 3 November 1972. Part number M28787/129-1 supersedes part numbers 2658639 and M28787/129. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits of table I.

Maximum power dissipation

Circuit	50 mW at 1 A.
Module	1 W.

Input current

Pin	1 A maximum.
Module	20 A maximum.

Input voltage

Pin	40 V maximum.

6.5 Module function. The module consists of 20 jumper lines whose purpose is to provide a means of isolating an operative circuit or subsystem from one that is inoperative. Removal of the EMF module makes possible complete isolation between two circuits or two subsystems.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N247)

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TABLE II. Module continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Module continuity	1	20	0	2	Ω	2

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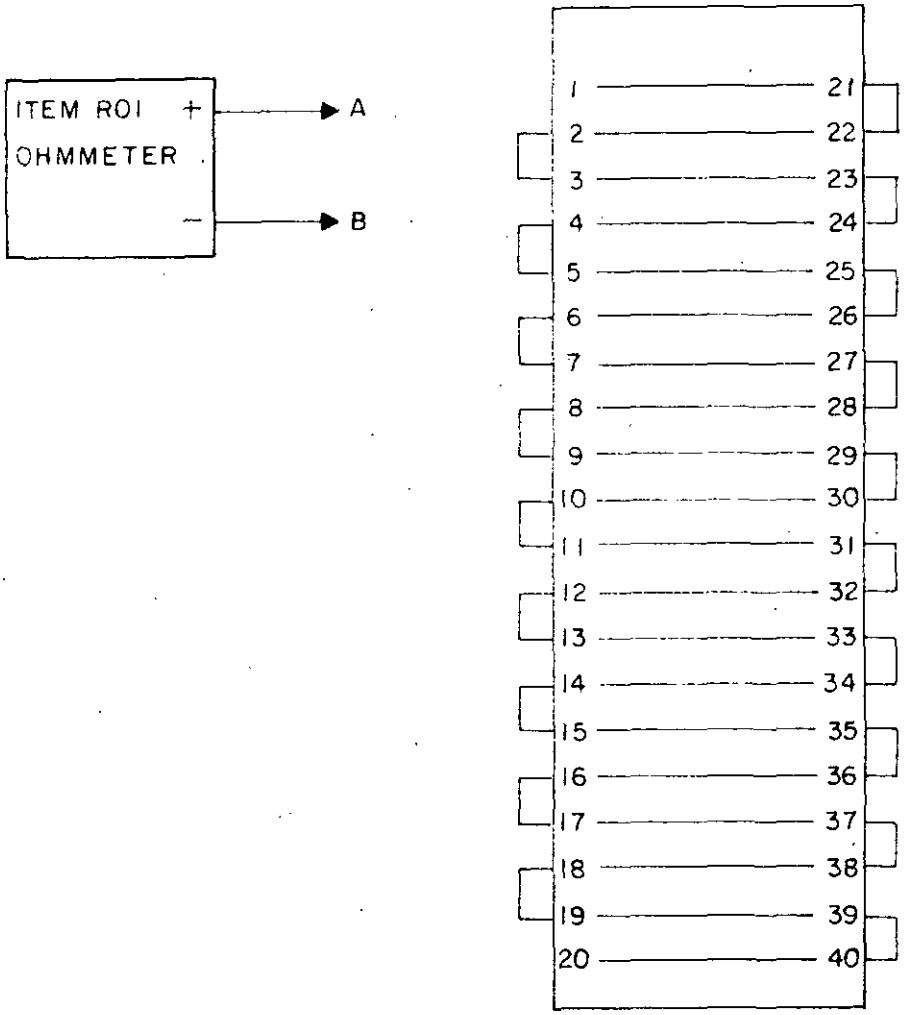


FIGURE 2. Module continuity test circuit for table II.

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TABLE III. Module integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Module integrity	1	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20	10	∞	MΩ	3
B2		2	1,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
B3		3	1,2,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
B4		4	1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
B5		5	1,2,3,4,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
B6		6	1,2,3,4,5,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
B7		7	1,2,3,4,5,6,8,9,10,11,12,13,14,15,16,17,18,19,20				
B8		8	1,2,3,4,5,6,7,9,10,11,12,13,14,15,16,17,18,19,20				

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TABLE III. Module integrity test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B9	Module integrity	9	1,2,3,4,5,6,7,8,10,11,12,13,14,15,16,17,18,19,20	10	∞	MΩ	3
B10		10	1,2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20				
B11		11	1,2,3,4,5,6,7,8,9,10,12,13,14,15,16,17,18,19,20				
B12		12	1,2,3,4,5,6,7,8,9,10,11,13,14,15,16,17,18,19,20				
B13		13	1,2,3,4,5,6,7,8,9,10,11,12,14,15,16,17,18,19,20				
B14		14	1,2,3,4,5,6,7,8,9,10,11,12,13,15,16,17,18,19,20				
B15		15	1,2,3,4,5,6,7,8,9,10,11,12,13,14,16,17,18,19,20				
B16		16	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,17,18,19,20				

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TABLE III. Module integrity test - Continued.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B17	Module integrity	17	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,18,19,20	10	∞	MΩ	3
B18	↓	18	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,19,20	↓	↓	↓	↓
B19		19	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,20				
B20		20	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,19				

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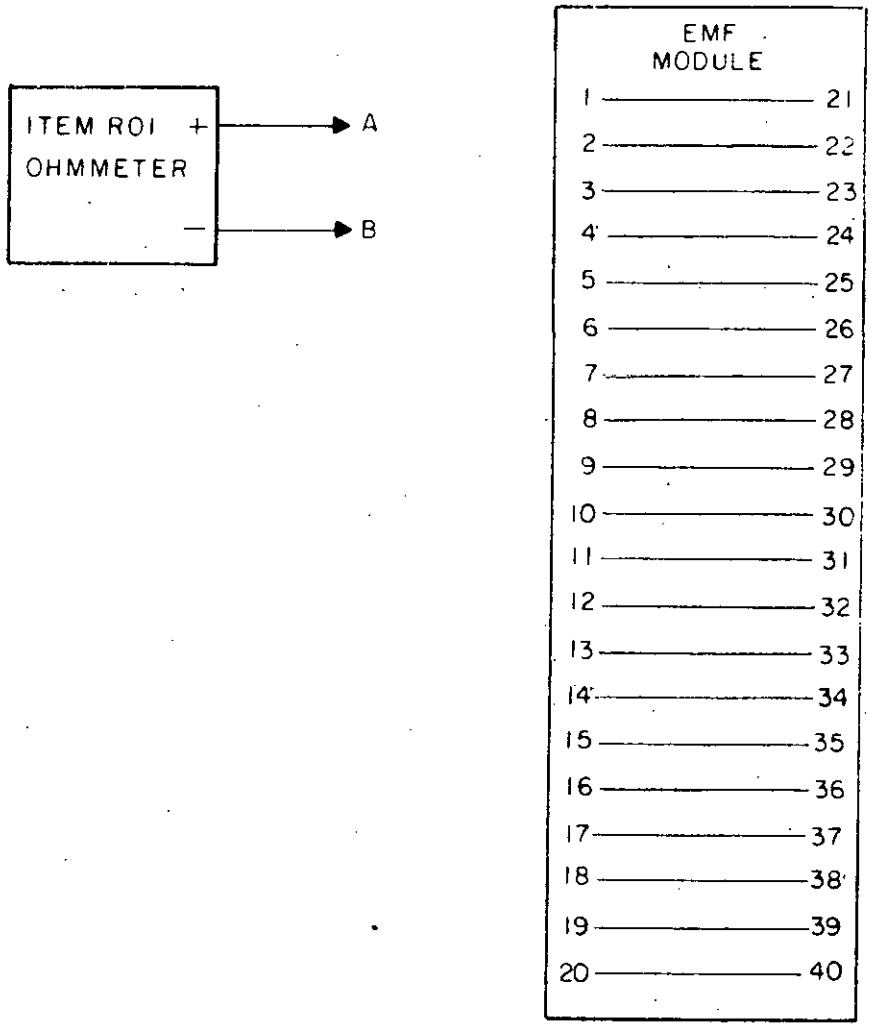


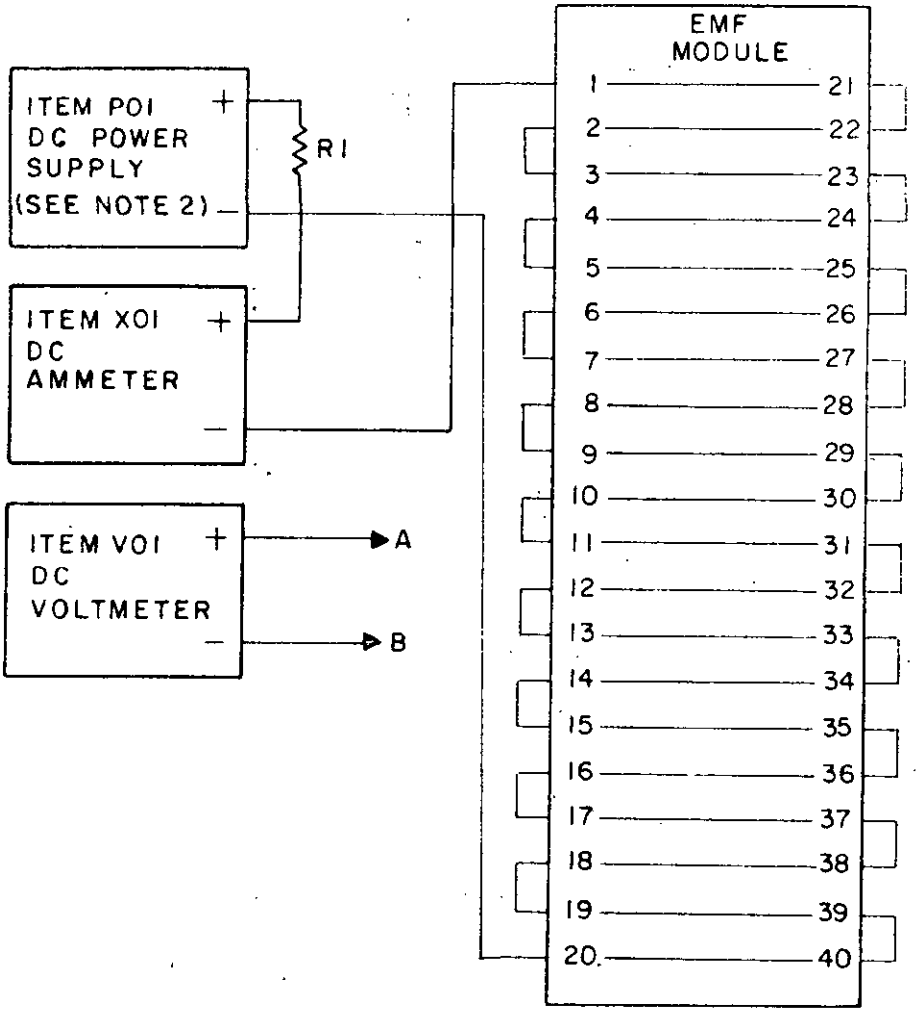
FIGURE 3. Module integrity test circuit for table III.

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TABLE IV. Jumper resistance check test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Jumper resistance check	1	21	0	50	mV	4
C2		3	23				
C3		5	25				
C4		7	27				
C5		9	29				
C6		11	31				
C7		13	33				
C8		15	35				
C9		17	37				
C10		19	39				
C11		22	2				
C12		24	4				
C13		26	6				
C14		28	8				
C15		30	10				
C16		32	12				
C17		34	14				
C18		36	16				
C19		38	18				
C20		40	20				

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NOTES:
1. R1 = 10Ω ±5%, 15W.
2. Adjust item P01 so that item X01 reads 1 ampere. The power supply voltage will be approximately 10V.

CAUTION
DO NOT ISOLATE THE CHASSIS OF AMMETER X01 FROM EARTH GROUND.

FIGURE 4. Jumper resistance check test circuit for table IV.

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TABLE V. *Insulation resistance test.*

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Insulation resistance	1	2,3,4,5,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20	10	∞	MΩ	5
D2		2	1,3,4,5,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D3		3	1,2,4,5,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D4		4	1,2,3,5,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D5		5	1,2,3,4,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D6		6	1,2,3,4,5,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D7		7	1,2,3,4,5,6, 8,9,10,11,12, 13,14,15,16, 17,18,19,20				
D8		8	1,2,3,4,5,6, 7,9,10,11,12, 13,14,15,16, 17,18,19,20				

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TABLE V. Insulation resistance test - Continued.

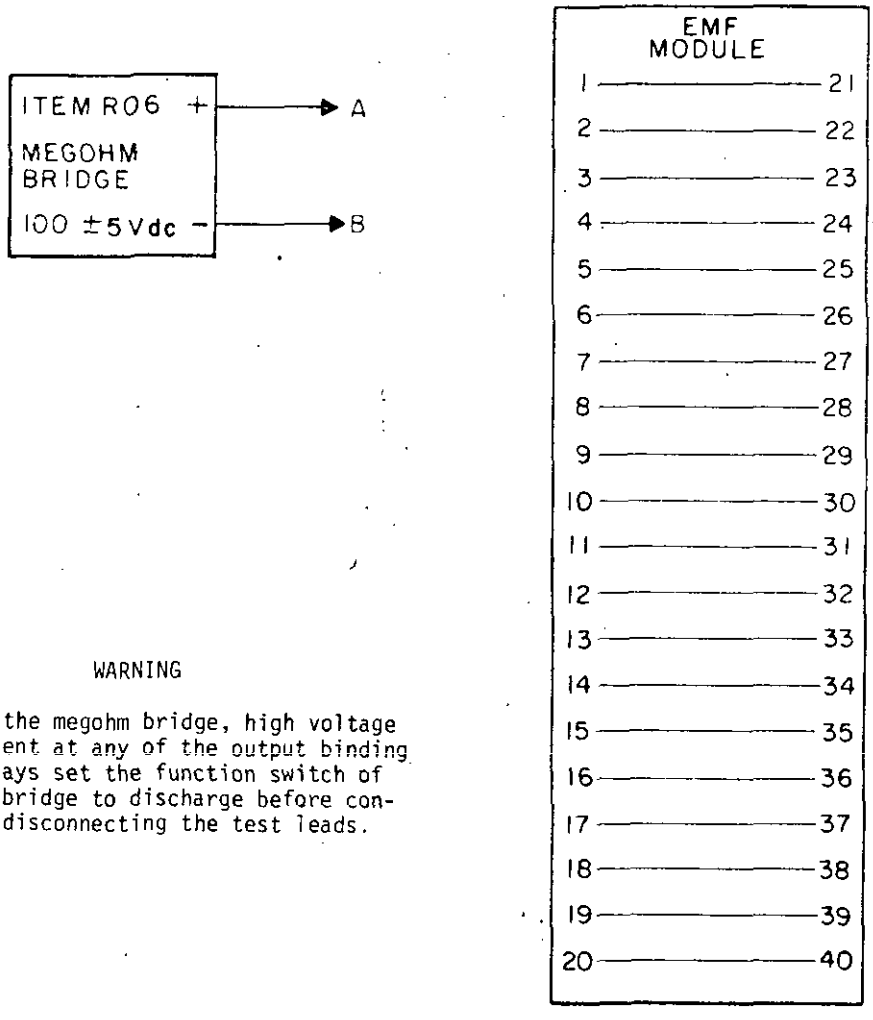
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D9	Insulation resistance	9	1,2,3,4,5,6, 7,8,10,11,12, 13,14,15,16, 17,18,19,20	10	∞	MΩ	5
D10		10	1,2,3,4,5,6, 7,8,9,11,12, 13,14,15,16, 17,18,19,20				
D11		11	1,2,3,4,5,6, 7,8,9,10,12, 13,14,15,16, 17,18,19,20				
D12		12	1,2,3,4,5,6, 7,8,9,10,11, 13,14,15,16, 17,18,19,20				
D13		13	1,2,3,4,5,6, 7,8,9,10,11, 12,14,15,16, 17,18,19,20				
D14		14	1,2,3,4,5,6, 7,8,9,10,11, 12,13,15,16, 17,18,19,20				
D15		15	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,16, 17,18,19,20				
D16		16	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 17,18,19,20				

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TABLE V. Insulation resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D17	Insulation resistance	17	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,18,19,20	10	∞	MΩ	5
D18		18	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,19,20				
D19		19	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,20				
D20		20	1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,19				

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads.

FIGURE 5. Insulation resistance test circuit for table V.

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TABLE VI. Capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Capacitance	1	2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20	0	10	pF	6
E2		2	1,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
E3		3	1,2,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
E4		4	1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
E5		5	1,2,3,4,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
E6		6	1,2,3,4,5,7,8,9,10,11,12,13,14,15,16,17,18,19,20				
E7		7	1,2,3,4,5,6,8,9,10,11,12,13,14,15,16,17,18,19,20				
E8		8	1,2,3,4,5,6,7,9,10,11,12,13,14,15,16,17,18,19,20				

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TABLE VI. Capacitance test - Continued.

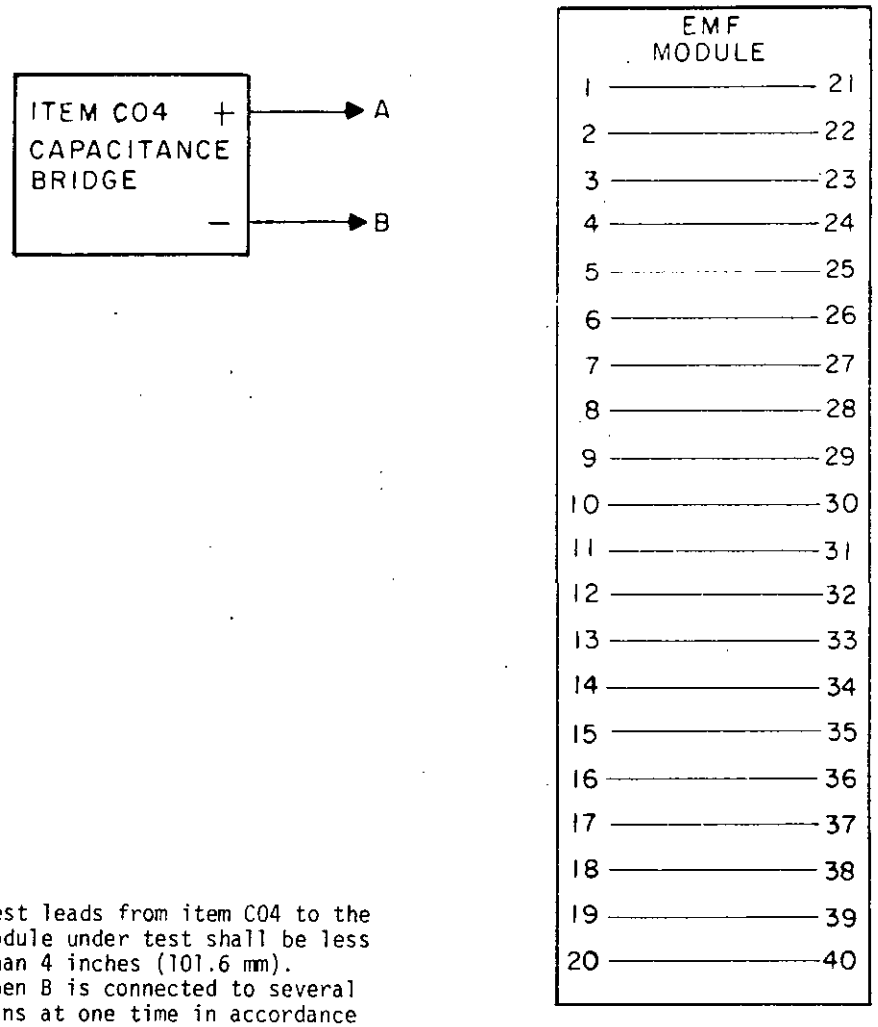
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E9	Capacitance	9	1,2,3,4,5,6,7,8,10,11,12,13,14,15,16,17,18,19,20	0	10	pF	6
E10		10	1,2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20				
E11		11	1,2,3,4,5,6,7,8,9,10,12,13,14,15,16,17,18,19,20				
E12		12	1,2,3,4,5,6,7,8,9,10,11,13,14,15,16,17,18,19,20				
E13		13	1,2,3,4,5,6,7,8,9,10,11,12,14,15,16,17,18,19,20				
E14		14	1,2,3,4,5,6,7,8,9,10,11,12,13,15,16,17,18,19,20				
E15		15	1,2,3,4,5,6,7,8,9,10,11,12,13,14,16,17,18,19,20				
E16		16	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,17,18,19,20				

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TABLE VI. Capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E17	Capacitance	17	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,18,19,20	0	10	pF	6
E18		18	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,19,20				
E19		19	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,20				
E20		20	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19				

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NOTES:
1. Test leads from item C04 to the module under test shall be less than 4 inches (101.6 mm).
2. When B is connected to several pins at one time in accordance with Table VI, the wire interconnecting the pins shall be less than 1 inch (25.4 mm) between each set of pins.

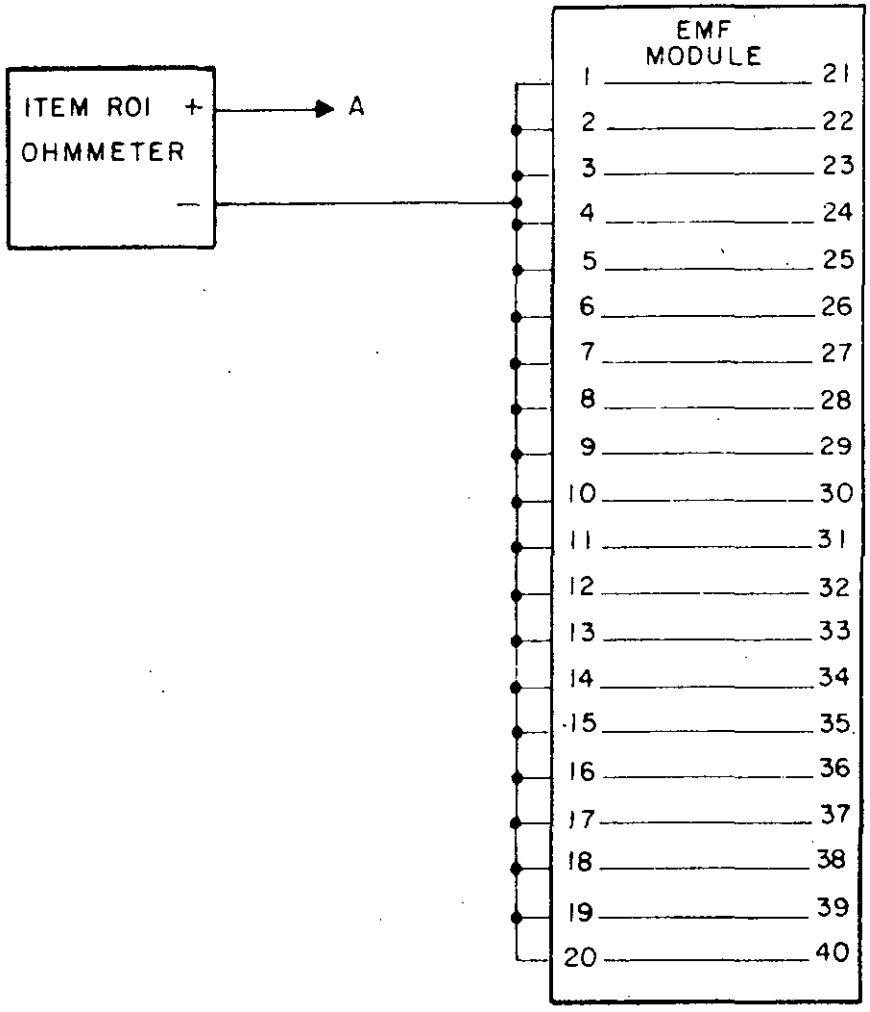
FIGURE 6. Capacitance test circuit for table VI.

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TABLE VII. Frame integrity test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F	Frame integrity	Frame	10	∞	MΩ	7

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NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the positive terminal.

FIGURE 7. Frame integrity test circuit for table VII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/129A(EC) - Modules, Standard Electronic Isolation Module, Digital, Key Code EMF

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